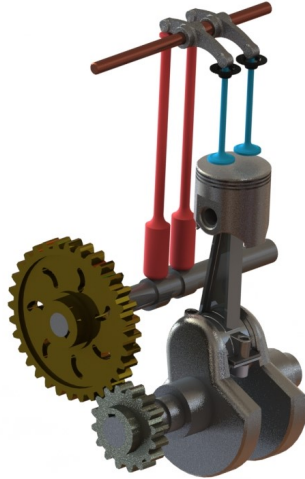




Design and Mechanical Analysis of a Pushrod (OHV) Valve Train Mechanism

COURSE: INTERNAL COMBUSTION ENGINES I

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Abstract

This report presents the design, kinematic analysis, and prototyping of an Overhead Valve (OHV) pushrod mechanism for a 328 cc naturally aspirated engine. The primary objective is to evaluate the structural integrity and motion transmission efficiency of the valve train. The project involves detailed stress calculations for critical components including the camshaft, pushrod, crankshaft, and piston, using a comparative approach between high-strength engineering materials (Steel) and 3D-printed polymers (PLA/ABS) used in the physical prototype. The results validate the functional design of the assembly, providing a safety factor analysis that ensures the prototype's operational reliability while highlighting the material performance gaps in real-world engine environments.

1 Introduction

The internal combustion engine's efficiency and performance are heavily dependent on the precision of the valve timing mechanism. Among the various designs, the Overhead Valve (OHV) system, commonly known as the "Pushrod" engine, remains a fundamental study in mechanical engineering due to its unique compact design and reliability. This mechanism utilizes a camshaft situated within the engine block, which actuates the valves via pushrods and rocker arms.

The primary goal of this project is to design, analyze, and prototype a functional 3D-printed model of a Pushrod valve train. The study focuses on understanding the kinematic relationship between the camshaft motion and the valve lift, ensuring that the mechanical linkages can withstand the operational forces while maintaining precise timing. This report bridges the gap between theoretical calculations—covering component stresses and engine power metrics—and the physical realization of the assembly.

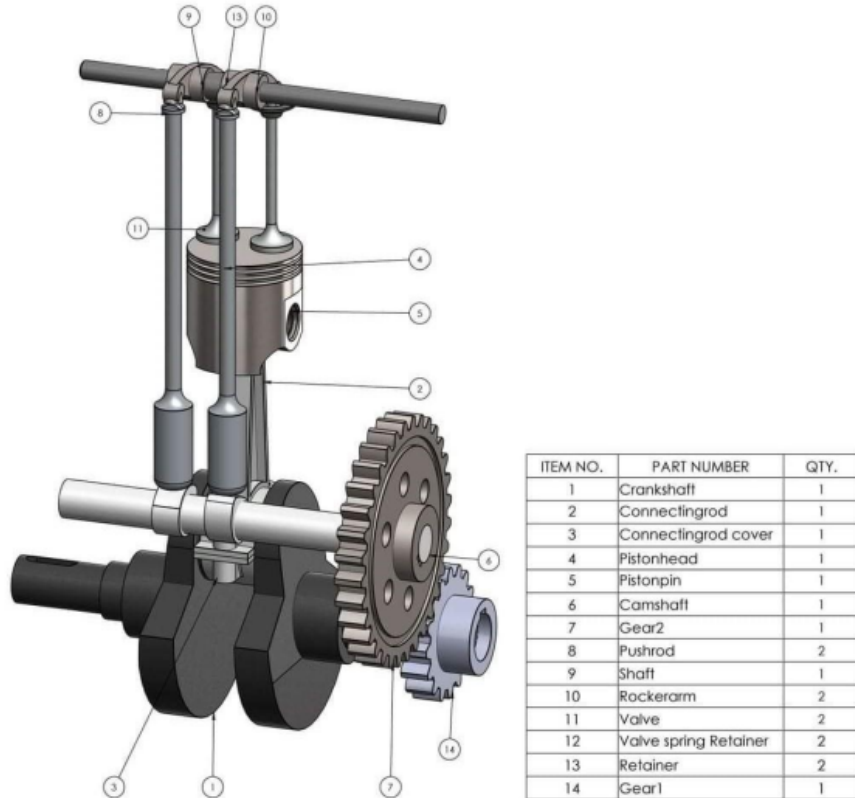


Figure 1: our Design model on SOLIDWORKS

1.1 Material Analysis and Comparative Study

A distinguishing feature of this project is the transition from a digital design to a physical 3D-printed prototype. While the theoretical calculations assume high-strength industrial materials to withstand peak combustion pressures and mechanical loads, the physical model is manufactured using Additive Manufacturing (3D Printing).

To provide a comprehensive formal report, we compare the properties of the material used in the prototype with standard engine materials:

Table 1: Comparative Material Properties: Prototype vs. Real-world Engine

Property	3D Printed (PLA/ABS)	Forged Steel / Alloy	Unit
Yield Strength (σ_y)	30 – 50	478 – 600	MPa
Young’s Modulus (E)	2 – 3	200 – 210	GPa
Density (ρ)	1.04 – 1.25	7.8 – 8.0	g/cm^3
Application	Functional Prototyping	High-Performance ICE	-

The following sections detail the mechanical stresses calculated based on a yield strength of 478 MPa (representing the industrial standard) while acknowledging the safety factor adjustments required for the 3D-printed assembly.

2 Design Analysis and Stress Calculations

To ensure the functional integrity of the valve train model, a rigorous mechanical analysis was performed. This section details the mathematical approach used to determine the critical parameters of the system. The calculations cover the cam profile geometry, and the stress distribution across the camshaft, pushrod, and piston assembly. By applying fundamental principles of machine design and kinematics, these calculations provide the theoretical foundation for the physical prototype, ensuring that the motion is transmitted accurately from the camshaft to the valve assembly.

So here a formal mechanical stress assessment of the main valve-train components under two material conditions:

- **Prototype material (3D-printed PLA):** $\sigma_y = 40$ MPa, $E = 2.5$ GPa = 2500 MPa.
- **Industrial material (forged steel):** $\sigma_y = 478$ MPa, $E = 210$ GPa = 210000 MPa.

All calculations are performed using consistent **N–mm–MPa** units, where $1 \text{ MPa} = 1 \text{ N/mm}^2$.

2.1 Stress Definitions

Several stress modes may act on the mechanical components of the system. **Normal stress** represents the internal resistance developed in a member subjected to axial loading. **Bending stress** occurs when a component is subjected to a moment that causes one side to go into tension and the other into compression. **Torsional shear stress** develops when torque is transmitted through a shaft. **Buckling** is an instability failure mode that may occur in slender members under compressive loading before the material reaches its compressive strength.

For the piston, two additional stress modes are especially important as mentioned in [2]. **Hoop stress** is the circumferential tensile stress induced in the piston skirt or cylindrical wall due to internal pressure. **Crown stress** refers to the bending stress developed in the piston crown as a thin plate subjected to combustion pressure.

2.2 Camshaft Analysis: Combined Bending and Torsion

The camshaft is idealized as a solid circular shaft with diameter $D = 20$ mm, subjected to a bending moment $M = 500$ N mm and a torque $T = 300$ N mm. as in [4]

The second moment of area and polar moment of inertia are:

$$I = \frac{\pi D^4}{64} = \frac{\pi(20)^4}{64} = 7853.98 \text{ mm}^4, \quad (1)$$

$$J = \frac{\pi D^4}{32} = \frac{\pi(20)^4}{32} = 15707.96 \text{ mm}^4 \quad (2)$$

The bending stress and torsional shear stress are:

$$\sigma_b = \frac{32M}{\pi D^3} = \frac{32(500)}{\pi(20)^3} = 0.6366 \text{ MPa}, \quad (3)$$

$$\tau = \frac{16T}{\pi D^3} = \frac{16(300)}{\pi(20)^3} = 0.1910 \text{ MPa} \quad (4)$$

The equivalent von Mises stress is therefore:

$$\sigma_{vM} = \sqrt{\sigma_b^2 + 3\tau^2} = \sqrt{(0.6366)^2 + 3(0.1910)^2} = 0.7178 \text{ MPa} \quad (5)$$

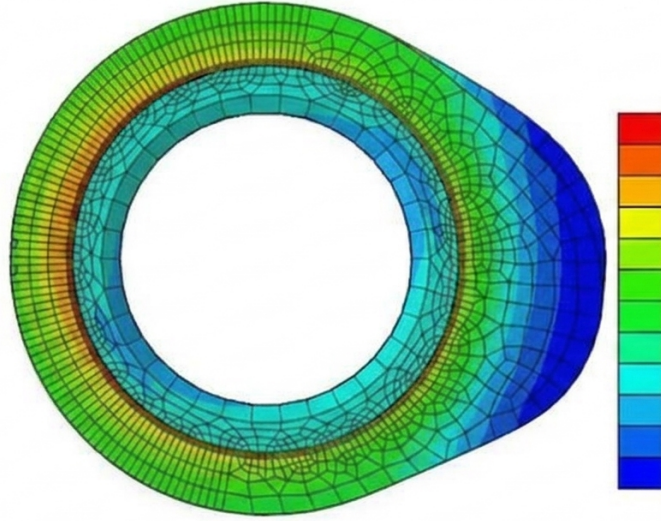


Figure 2: Illustration of vonmises stress distribution of the cam from [10]

Safety factor assessment:

$$n_s = \frac{478}{0.7178} = 665.9, \quad (6)$$

$$n_p = \frac{40}{0.7178} = 55.7 \quad (7)$$

2.3 Pushrod Analysis: Axial Compression and Buckling

The pushrod is modeled as a solid circular column with diameter $D = 20$ mm and length $L = 200$ mm, carrying an axial compressive load $F = 2000$ N as mentioned in [4].

2.3.1 Axial Stress

The cross-sectional area is:

$$A = \frac{\pi D^2}{4} = \frac{\pi(20)^2}{4} = 314.16 \text{ mm}^2 \quad (8)$$

Thus, the axial stress is:

$$\sigma_a = \frac{F}{A} = \frac{2000}{314.16} = 6.366 \text{ MPa} \quad (9)$$

Axial safety factor:

$$n_{s,ax} = \frac{478}{6.366} = 75.06, \quad (10)$$

$$n_{p,ax} = \frac{40}{6.366} = 6.28 \quad (11)$$

2.3.2 Euler Buckling Analysis

For a solid circular section:

$$I = \frac{\pi D^4}{64} = \frac{\pi(20)^4}{64} = 7853.98 \text{ mm}^4 \quad (12)$$

Assuming pinned-pinned end conditions, $K = 1$, the critical buckling load is:

$$P_{cr,s} = \frac{\pi^2 E_s I}{(KL)^2} = \frac{\pi^2 (210000)(7853.98)}{(2000)^2} = 4.0696 \times 10^3 \text{ N}, \quad (13)$$

$$P_{cr,p} = \frac{\pi^2 E_p I}{(KL)^2} = \frac{\pi^2 (2500)(7853.98)}{(2000)^2} = 4844.7 \text{ N} \quad (14)$$

Buckling safety factor:

$$n_{buck,s} = \frac{P_{cr,s}}{F} = \frac{4.0696 \times 10^3}{2000} = 20.348, \quad (15)$$

$$n_{buck,p} = \frac{P_{cr,p}}{F} = \frac{4844.7}{2000} = 2.42 \quad (16)$$

2.4 Crankshaft Analysis: Combined Bending and Torsion

The crankshaft is idealized as a solid circular shaft with $D = 25 \text{ mm}$, subjected to a bending moment $M = 5000 \text{ N mm}$ and a torque $T = 3000 \text{ N mm}$.

The bending and torsional stresses are:

$$\sigma_b = \frac{32M}{\pi D^3} = \frac{32(5000)}{\pi(25)^3} = 3.26 \text{ MPa}, \quad (17)$$

$$\tau = \frac{16T}{\pi D^3} = \frac{16(3000)}{\pi(25)^3} = 0.98 \text{ MPa} \quad (18)$$

The equivalent von Mises stress is:

$$\sigma_{vM} = \sqrt{\sigma_b^2 + 3\tau^2} = \sqrt{(3.26)^2 + 3(0.98)^2} = 3.6 \text{ MPa} \quad (19)$$

Safety factor assessment:

$$n_s = \frac{478}{3.6} = 133 \quad (20)$$

$$n_p = \frac{40}{0.98} = 63.5 \quad (21)$$

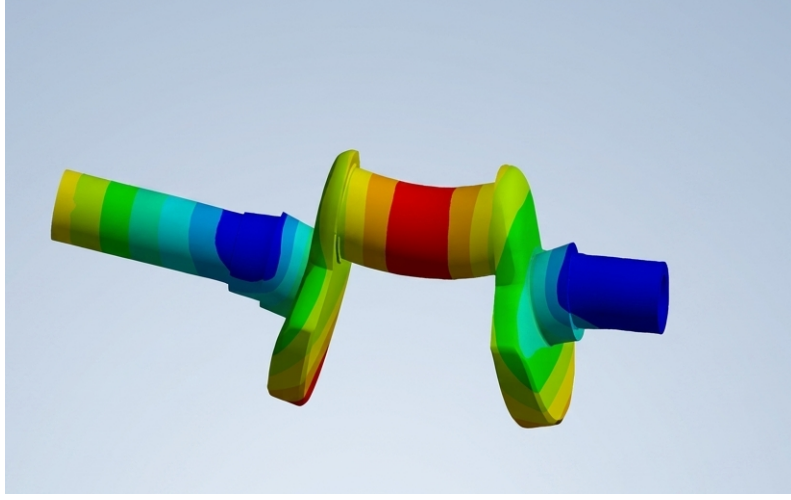


Figure 3: Loads impact on the crankshaft.

2.5 Piston Analysis: Axial, Hoop, and Crown Stresses

The piston is modeled with an outer diameter $D = 60$ mm, combustion pressure $p = 5$ MPa, and crown thickness $t = 5$ mm. Noticing that the material is Aliminuim Alloy used much more in pistons manufacturing with yield strength = 150 MPa.

2.5.1 Gas Force on the Piston Crown

The effective piston area is:

$$A = \frac{\pi D^2}{4} = \frac{\pi(60)^2}{4} = 2827.43 \text{ mm}^2 \quad (22)$$

Thus, the total gas force acting on the piston is:

$$F_g = pA = 5 \times 2827.43 = 14137.17 \text{ N} \quad (23)$$

2.5.2 Direct Compressive Stress

The direct compressive stress transmitted by the combustion pressure is:

$$\sigma_{ax} = p = 5 \text{ MPa} \quad (24)$$

Safety factor:

$$n_{s,ax} = \frac{150}{5} = 30, \quad (25)$$

$$n_{p,ax} = \frac{40}{5} = 8.0 \quad (26)$$

2.5.3 Hoop Stress

As discussed in [1] The hoop stress in the piston wall is:

$$\sigma_h = \frac{pD}{2t} = \frac{5(60)}{2(1.5)} = 100 \text{ MPa} \quad (27)$$

Safety factor:

$$n_{s,h} = \frac{150}{100} = 1.5, \quad (28)$$

$$n_{p,h} = \frac{40}{100} = 0.4 \quad (29)$$

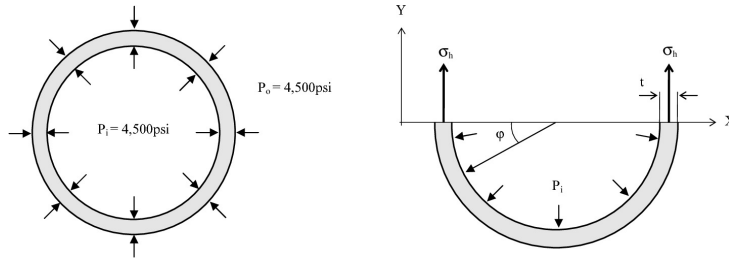


Figure 4: Illustration of hoop stress distribution according to [5] in a thin-walled cylindrical section subjected to internal pressure. The circumferential tensile stress develops along the piston wall due to combustion loading.

2.6 Summary of Structural Performance

Component	Stress / Load Mode	Value	FOS (Steel)	FOS (PLA)
Camshaft	Von Mises stress	0.718 MPa	665.9	55.7
Pushrod	Axial stress	6.37 MPa	75.06	6.28
Pushrod	Euler buckling load	$4.0696 \times 10^3 \text{ N}$	20.348	2.42
Crankshaft	Von Mises stress	0.630 MPa	133	63.5
Piston	Direct compressive stress	5 MPa	30	8.0
Piston	Hoop stress	30 MPa	1.5	0.4

Table 2: Computed safety factors comparison for the main components

2.7 Stress results

AS a result from the stresses calculations all the components are structurally safe under the assumed loading conditions for both materials. For the piston, the governing stress mode is the crown bending stress, followed by hoop stress. With thickness of 1.5 mm, the steel piston remains safe under the assumed pressure level, while the PLA piston is still not fully reliable for real combustion loading. Accordingly, the 3D-printed prototype is suitable for kinematic validation and low-load demonstration, whereas a functional engine application requires a stronger material and proper design optimization of the piston crown and skirt geometry.

3 Engine Calculations

The geometric configuration of the internal combustion engine defines its thermodynamic potential and volumetric efficiency. Based on the mechanical design principles outlined in [6], this section details the fundamental dimensions and volume calculations derived directly from the project drawings.

3.1 Given Dimensions from Drawings

The following parameters are established from the physical assembly and technical drawings of our pushrod engine model:

Parameter	Value
Bore (D)	60 mm
Crank Radius (r)	58 mm
Stroke ($L = 2r$)	116 mm
Compression Ratio (CR)	9:1

Table 3: Our Geometric Parameters

3.2 Volumetric Analysis

Using the primary dimensions, the engine capacity is calculated through the following volumetric stages:

3.2.1 Swept Volume (V_s)

The swept volume, or displacement, represents the volume through which the piston travels during a single stroke:

$$V_s = \frac{\pi}{4} \cdot D^2 \cdot L \quad (30)$$

Substituting the given values:

$$V_s = \frac{\pi}{4} \cdot (60)^2 \cdot 116 \approx 327,982 \text{ mm}^3 \approx 328 \text{ cc} \quad (31)$$

3.2.2 Clearance Volume (V_c)

The clearance volume is the space remaining in the cylinder when the piston is at Top Dead Center (TDC). It is derived by rearranging the compression ratio formula:

$$V_c = \frac{V_s}{CR - 1} \quad (32)$$

Substituting the values:

$$V_c = \frac{328}{9 - 1} = \frac{328}{8} = 41 \text{ cc} \quad (33)$$

3.2.3 Total Volume (V_{total})

The total cylinder volume is the sum of the swept and clearance volumes:

$$V_{total} = V_s + V_c \quad (34)$$

Substituting the values:

$$V_{total} = 328 + 41 = 369 \text{ cc} \quad (35)$$

3.3 Power and Mechanical Efficiency Analysis

Evaluating the power distribution is essential to understanding the mechanical losses within the pushrod assembly. The relationship between the power generated in the cylinder and the power delivered to the crankshaft is governed by the mechanical efficiency of the system.

3.3.1 Friction Power Formula

The friction power (FP) represents the total energy dissipated due to mechanical friction between moving parts, including the valve train and bearings. It is defined by the following relation:

$$FP = IP - BP \quad (36)$$

Where:

- FP : Friction Power.
- IP : Indicated Power (The total power generated in the combustion chamber).
- BP : Brake Power (The useful power available at the output shaft).

3.3.2 Estimation of Brake Power (BP)

For a 328 cc naturally aspirated push-rod engine, typical brake power values range between 5 kW and 10 kW [6]. For this analysis, a mid-range estimate is selected:

$$BP_{estimated} = 7.0 \text{ kW} \quad (37)$$

3.3.3 Calculation of Indicated Power (IP)

Assuming a standard mechanical efficiency (η_m) of 80% (0.8) for this class of engines, the Indicated Power is calculated by rearranging the efficiency equation:

$$IP = \frac{BP}{\eta_m} \quad (38)$$

Substituting the estimated values:

$$IP = \frac{7.0}{0.8} = 8.75 \text{ kW} \quad (39)$$

3.3.4 Calculation of Friction Power (FP)

Using the results from the previous steps, the power lost to friction is:

$$FP = IP - BP = 8.75 - 7.0 = 1.75 \text{ kW} \quad (40)$$

3.4 Summary of Performance Results

The following table summarizes the estimated performance characteristics of the designed engine:

3.5 Scavenging Ratio Analysis

The scavenging ratio (SR) is a critical parameter in internal combustion engines that defines the effectiveness of the gas exchange process. It represents the ratio of the mass of fresh air delivered to the cylinder to the mass of air that would fill the displacement volume at ambient conditions.

Parameter	Estimated Value
Brake Power (BP)	7.0 kW
Mechanical Efficiency (η_m)	80%
Indicated Power (IP)	8.75 kW
Friction Power (FP)	1.75 kW

Table 4: Final Estimated Performance Metrics

3.5.1 Typical Scavenging Ratio Values

Based on standard engine design benchmarks [6], the scavenging ratio varies significantly depending on the engine type and induction system. Table 5 outlines the typical ranges used in industrial applications.

Engine Type	Typical SR Range
Small naturally aspirated engine	0.8 – 0.95
High-performance naturally aspirated engine	1.0 – 1.1
Supercharged two-stroke engine	1.2 – 1.8

Table 5: Typical Scavenging Ratio (SR) Ranges

3.5.2 Selected Scavenging Ratio

For the current 328 cc pushrod engine model, which operates as a small naturally aspirated unit, a scavenging ratio of approximately **0.9** is selected:

$$SR \approx 0.9 \quad (41)$$

This value indicates an efficient gas exchange process, ensuring that 90% of the cylinder’s displacement is effectively filled with fresh charge, which is consistent with the assumed mechanical efficiency of 80% and the calculated brake power of 7 kW.

4 MATLAB Simulation

This section presents the MATLAB/Simulink simulation of the internal combustion engine model used in this work. The simulation framework is based on the official MathWorks Generic Engine block documentation, which provides a configurable internal combustion engine model for spark-ignition and diesel applications, driven either by torque or normalized throttle, with optional fuel-consumption output and speed-control features. [7]

The presented results are organized in a logical sequence, starting from the camshaft motion representation, followed by the Simulink implementation of the cam mechanism, and then the engine performance plots obtained under different operating conditions.

4.1 Engine Control and Operating Signals

4.2 Cylinder Performance and Combustion Characteristics

The following plots illustrate the thermodynamic and flow characteristics of the engine cylinders over two complete operating cycles. These results provide insight into the intake process, combustion behavior, cylinder pressure evolution, and generated torque.

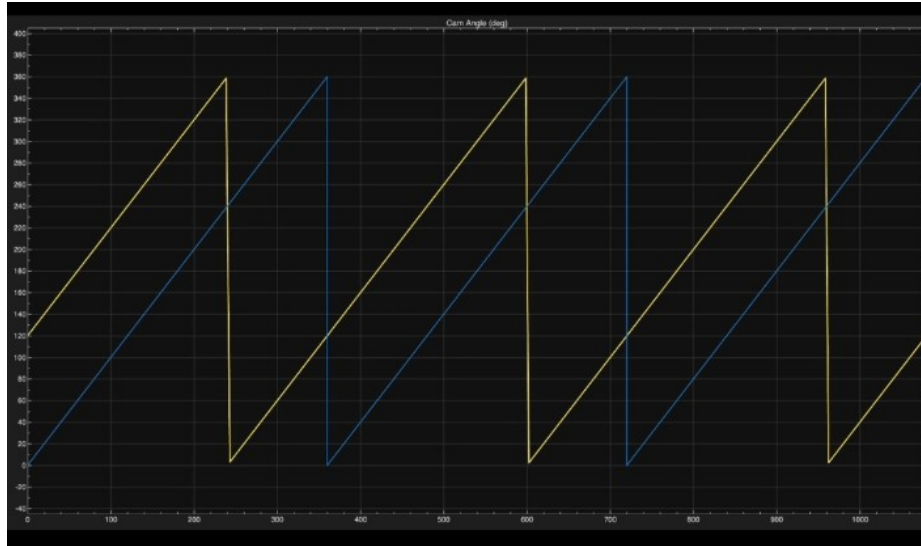


Figure 5: Camshaft rotation angle versus time.

```
open_system('Camshaft');

[theta_TLU, radius_TLU, length_TLU] = CamshaftGenerateCamProfile(radius_min, radius_max,
theta_constant, eccentricity);
CamshaftPlotCamProfile;

open_system('Camshaft/Cam Angle (deg)');
sim('Camshaft');
```

Figure 6: MATLAB/Simulink code snippet used to open the camshaft model in Simulink.

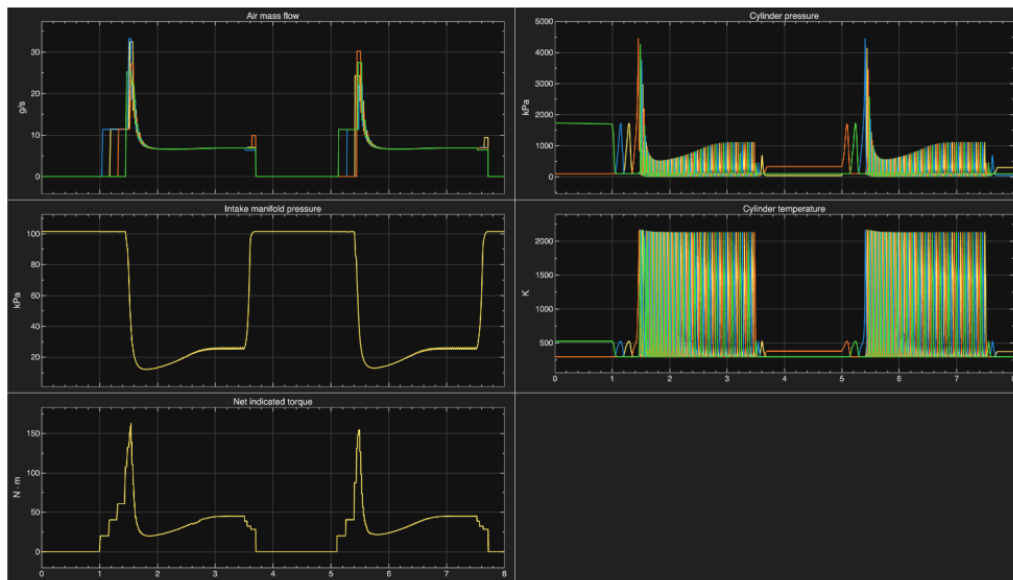


Figure 9: Engine performance parameters over two operating cycles, including air mass flow rate per cylinder, cylinder pressure variation, intake manifold pressure, cylinder temperature distribution, and net indicated torque, illustrating the coupled thermodynamic and mechanical behavior of the engine system.

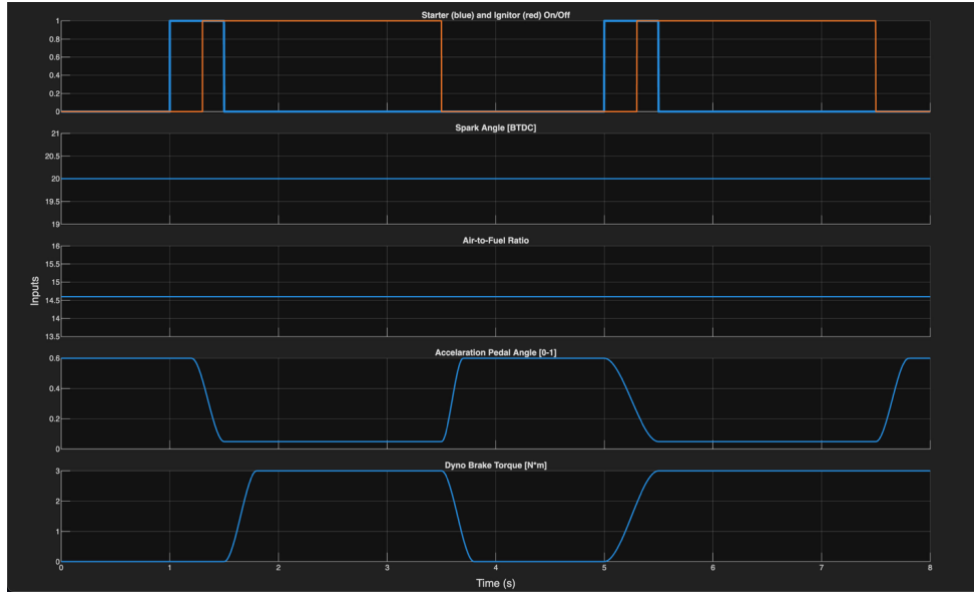


Figure 8: The Graph showing when the starter (blue) and ignitor (red) turn on or off. Second graph: Graph showing the spark angle value. Third graph: Graph showing the air-to-fuel ratio value. Fourth graph (didn't take it): Graph showing the acceleration pedal angle. Fifth graph: Graph showing the dyno brake torque.

4.4 Engine Speed and Torque Analysis

The following plots compare the instantaneous engine behavior with cycle-averaged values, highlighting speed fluctuations and torque interactions during operation.

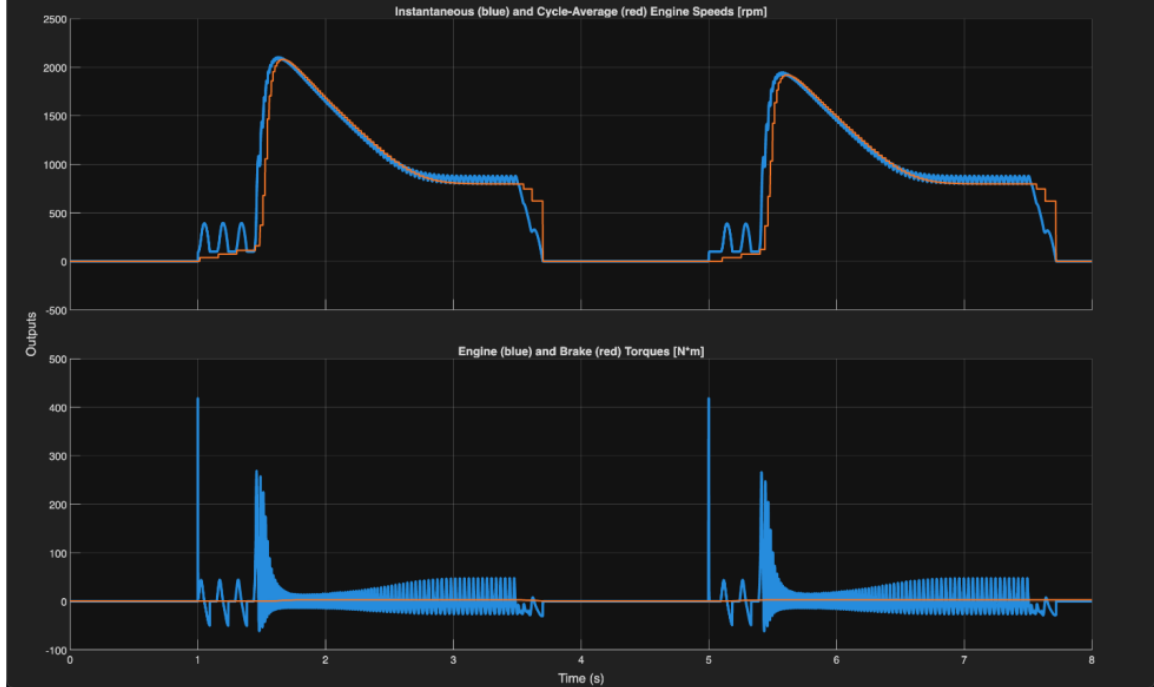


Figure 11: First graph comparing instantaneous engine speed (blue) with cycle-average engine speed (red) across 2 cycles. Second graph comparing engine torque (blue) with brake torque (red) across 2 cycles.

4.5 Complete Engine Simulink Model

The complete internal combustion engine model was implemented in Simulink using Simulink engine blocks and crank-angle-resolved subsystems. The model accepts multiple operating inputs such as brake torque, spark angle, and air-to-fuel ratio, then generates dynamic outputs including engine speed, torque, pressure, airflow, and combustion-related variables.

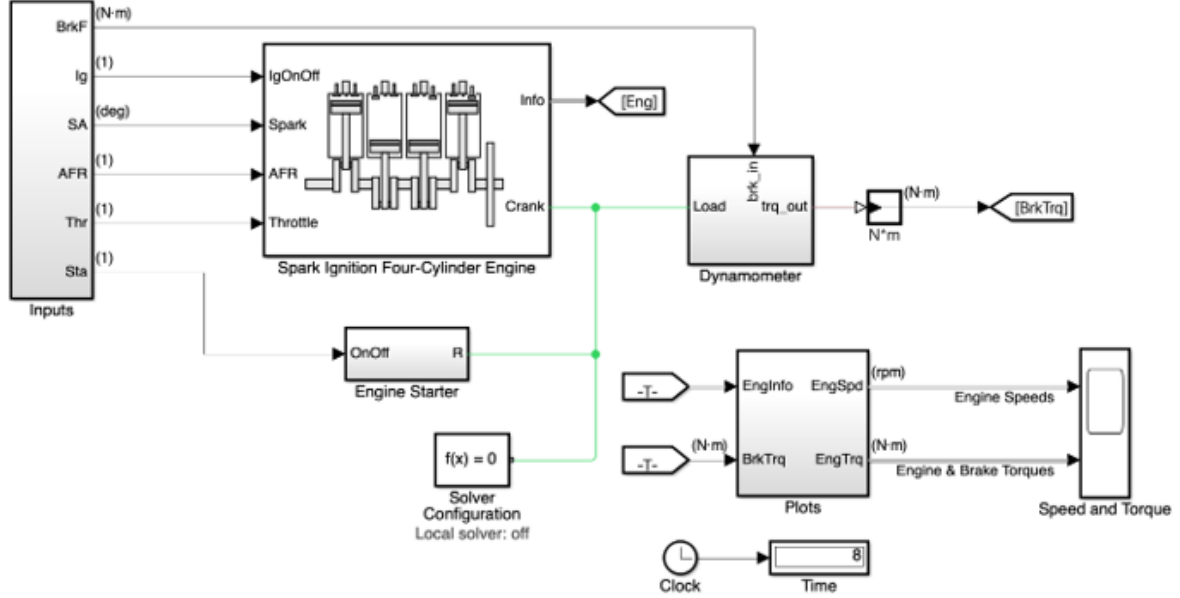


Figure 12: Complete Simulink model of the internal combustion engine system.

4.6 Simulink Engine Initialization Code

The following MATLAB/Simulink code initializes and opens the crank-angle-resolved engine model from Simulink and Symbolic Math Toolbox.

```
open_system('CrankAngleResolvedEngine')

CrankAngleResolvedEnginePlot;
```

Figure 13: MATLAB code snippet used to open the crank-angle-resolved engine model in Simulink.

4.7 Simulation Discussion

The MATLAB/Simulink simulations successfully demonstrated the dynamic behavior of the internal combustion engine model under different operating conditions. The obtained results showed realistic variations in cylinder pressure, air mass flow rate, combustion temperature, engine torque, and rotational speed throughout the engine cycles.

Additionally, the camshaft and crank-angle-resolved models provided a clear representation of valve timing and combustion synchronization. The simulation environment therefore served as an effective tool for validating the mechanical behavior and dynamic response of the proposed engine system before physical implementation.

5 Cost Analysis and Economic Comparison

This section compares the approximate market prices of the main components used in a conventional pushrod engine with the cost structure of the developed educational prototype. The objective is to highlight the economic difference between a full-scale industrial mechanism and a simplified demonstrator model, while also clarifying the main factors that affect component pricing in real applications.

5.1 Market Prices of Conventional Engine Components

Table 6 as mentioned in [8] summarizes the typical price ranges of the major engine components considered in this study. The values represent approximate market prices in Egyptian Pounds (EGP) and are intended to provide a realistic comparison between the industrial mechanism and the prototype system.

Part	Main Material(s)	Typical Price Range (EGP)
Camshaft	Hardened steel or cast iron	8,000 – 42,000
Lifters (Tappets)	Hardened steel	2,600 – 21,000
Pushrods	Chromoly steel or aluminum	1,000 – 10,500
Rocker Arms	Steel, aluminum, or forged alloy	2,600 – 26,000
Intake Valves	Stainless steel	1,000 – 3,200 each
Exhaust Valves	Heat-resistant stainless steel	1,600 – 4,200 each
Timing Gears	Steel or composite alloy	2,600 – 16,000
Crankshaft	Forged steel or cast iron	10,500 – 80,000
Pistons	Aluminum alloy	5,300 – 32,000 per set
Connecting Rods	Forged steel or aluminum	5,300 – 42,000 per set
Main Bearings	Steel-backed aluminum/copper	2,100 – 10,500

Table 6: Approximate market prices of conventional engine components in EGP

5.2 Why Engine Component Prices Vary

The cost of engine components depends on several practical and commercial factors. First, prices differ between OEM parts and aftermarket alternatives. Second, imported parts are often more expensive than locally available components. Third, the final cost depends on the engine type, since parts used in V6, V8, truck, or performance applications are not identical in design and manufacturing complexity. In addition, racing-grade components usually have higher prices due to tighter tolerances and stronger materials. Finally, the condition of the part, whether new or used, has a direct influence on the final market value.

5.3 Common Materials Used in Pushrod Engines

Conventional pushrod engines commonly use cast iron, aluminum alloys, forged steel, stainless steel, and chromoly steel. Cast iron is widely used in heavy-duty applications because of its durability and strength. Aluminum alloys are preferred when weight reduction and improved heat dissipation are important. Forged steel is selected for high-strength rotating or reciprocating parts. Stainless steel provides good corrosion and thermal resistance, while chromoly steel is often used in pushrods because it offers a favorable balance between strength and weight.

5.4 Economic cost of our Prototype

Compared with a full-scale industrial engine, the developed model is intended as an educational platform rather than a production-ready mechanism. Therefore, its material selection, manufacturing method, and overall component complexity are significantly simplified. Instead of forged or heat-treated metallic components, the prototype use 3D-printed parts, simplified shafts, and simple base, which substantially reduces the total fabrication cost.

This makes the prototype suitable for:

- demonstrating the operating principle of the valve train
- validating kinematic motion
- performing low-load functional testing

Table 7 is provided for presenting the cost of the developed model.

Prototype Component	Material / Item	Cost (EGP)
Camshaft model	3D-printed PLA / printed support parts	90
Pushrods	3D-printed PLA	40
Rocker arms	3D-printed plastic parts	60
Base frame	wood	80
Piston	3D-printed PLA	80
Crank shaft	3D-printed PLA	120
Connecting Rod	3D-printed PLA	80
Valve System	3D-printed PLA	50
Gears	3D-printed PLA	100
Total		700

Table 7: Prototype cost breakdown for the developed engine model.

6 Advantages and disadvantages of pushrod engine system

Pushrod System Advantages	Pushrod System Disadvantages
Strong low-end torque, especially at low engine speeds, due to the compact valvetrain layout and improved air velocity in low-RPM operation.	Limited high-RPM performance because the long pushrods and rocker arms increase valvetrain inertia, which can lead to valve float at very high speeds.
Simple and reliable design, with a compact camshaft layout and fewer moving parts compared with DOHC systems.	Less suitable for extreme performance applications, since the valvetrain geometry becomes more stressed as engine speed and boost increase.
Lighter and narrower engine packaging, which can improve center of gravity and leave more room for intake, exhaust, and other modifications.	More sensitive to mechanical stress under very high load, which may increase wear, fatigue, or the risk of valve-train failure.
Lower manufacturing cost and easier maintenance because the design is mechanically simpler than a DOHC setup.	Less flexibility for very aggressive valve timing and high-flow performance tuning compared with DOHC engines.

Table 8: Comparison of pushrod engine system advantages and disadvantages [9].

7 Conclusion

This report presented the comprehensive design, kinematic analysis, and prototyping of an Overhead Valve (OHV) pushrod mechanism for a 328 cc naturally aspirated engine. The primary objective to evaluate structural integrity and motion transmission efficiency was successfully achieved through a combination of theoretical calculations and physical model development.

The mechanical stress assessment of critical components—including the camshaft, pushrod, crankshaft, and piston—validated the functional design of the assembly. Comparative analysis between industrial forged steel and 3D-printed PLA revealed that while all components are structurally safe under assumed loading conditions, industrial materials remain necessary for real engine environments. Specifically, the 3D-printed PLA piston demonstrated a hoop stress safety factor of 0.4, indicating it is suitable for kinematic validation but not for actual combustion loading.

Engine calculations established the thermodynamic potential of the design, with a calculated swept volume of approximately 328 cc. Performance analysis estimated the brake power (BP) at 7.0 kW and the indicated power (IP) at 8.75 kW, based on a standard mechanical efficiency of 80%. These metrics were corroborated by MATLAB/Simulink simulations, which successfully demonstrated realistic dynamic behavior, including valve timing, cylinder pressure variations, and net indicated torque.

From an economic perspective, the project demonstrated that an effective educational platform can be realized at a fraction of industrial costs, with the prototype totaling 700 EGP. Ultimately, this study bridges the gap between theoretical machine design and physical realization, providing a functional model for demonstrating the operating principles of a pushrod valve train system.

8 Credits and Author Contributions

- **Eslam Abd El-Hamid:** SolidWorks design and final report preparation.
- **Mohamed Ahmed Mohamed:** Prototype fabrication and assembly, with contribution to the initial design concept.
- **Eslam Mossad:** Structural and stress analysis of the engine components.
- **Mohamed Amr:** Stress analysis and project presentation preparation.
- **Mohamed Karim:** MATLAB/Simulink modeling and simulations.
- **Marwan Ayman Serag El-Din:** Engine performance calculations.

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